

Junsu Lee

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EDUCATION

Oregon State University

Jan 2021-Jan 2024

Bachelors of Science in Computer Science

WORK EXPERIENCE

Production Engineer

Jan 2026 – Present

LG Energy Solution, ESS Battery Production | Holland, MI

- Designed and implemented temporal clustering algorithms to identify discrete production events from continuous machine sensor streams, enabling accurate computation of manufacturing KPIs that were previously impossible to measure directly from available data.
- Designed and implemented an automated Spotfire dashboard transforming raw manufacturing sensor data into real-time KPI visualizations, eliminating manual data processing and saving engineers 15 hours weekly.
- Prepared and led daily engineer meetings regarding production, downtime, and other operating information to establish daily projects and goals.
- Contributed to the increase of facility operation rate from 62% to 89% by resolving production bottlenecks and streamlining material recovery protocols.
- Evaluated hold material flagged by vision systems with specific recovery plans for each battery electrode.

Software Developer

Jan 2024 – Mar 2025

MMT Prep. LLC, Internal Tutoring Software | Portland, OR

- Created a PDF parsing pipeline using boundary detection and an LLM-driven state machine to accurately extract mathematical content from complex educational PDFs.
- Developed the question library application using SQLite, React, and Node.js to store 100,000 unique questions allowing tutors to read, edit, and export fully-rendered LaTeX text.
- Built dynamic question pipelines to generate topic-based math questions with step-by-step solutions to expand the product line.

PROJECTS

League of Legends Champion App | Python, PyTorch, SQL

- Designed and trained a PyTorch neural network that predicts match outcomes based on team compositions, achieving a 5.0% winrate edge over baseline using suggested picks. Implemented end-to-end data pipelines for 1.5 million ranked matches using embedding layers and regularization.

Koi Fish Simulation | Three.js, GLSL, React

- Developed an interactive 3D visualizer using Three.js and GLSL to use in portfolio website. Implemented the Boids flocking algorithm and custom shaders using noise functions for unique procedural patterning.

SKILLS

Languages: Python, JavaScript, SQL, C++

Frameworks: React, Node.js, PyTorch, Numpy

Tools: Docker, Git, Linux, SQLite, Spotfire

Coursework: Machine Learning & Data Mining, Data Structures, Operating Systems I & II

Spoken Languages: English (*Native*), Korean (*Fluent*), Spanish (*conversational*), Japanese (*basic*)